

No. of Functional Features Included in the Solution

Date	02 July 2024
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	Crop Recommendation	Suggests the best crop based on soil and weather values.	ML Prediction	Done	High
2	Soil Data Input	Accepts N, P, K, pH, temperature, humidity and rainfall values.	User Interface	Done	High
3	Prediction Model	Uses trained ML model to predict suitable crops.	Machine Learning	Done	High
4	User-Friendly Dashboard	Displays prediction results in a simple interface.	Frontend	Done	Medium
5	Flask Backend	Connects frontend with ML model.	Backend	Done	Medium
6	Data Validation	Checks user input before prediction.	Backend	Done	Medium
7	Result Display	Shows recommended crop instantly.	Frontend	Done	Medium
8	Future Recommendation Module	Can be extended with fertilizer suggestions.	Future Scope	Pending	Low

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	7
Core / Must-Have Features	5
Additional / Nice-to-Have Features	- Future fertilizer recommendation - Government scheme suggestions
Features Tested & Verified	- Crop prediction - User input validation - Flask integration - ML model loading - Result generation

Feature Category Breakdown:

S.No.	Category	Features in Category	Example Feature
1	User Interface (UI)	2	Soil Input Form, Result Page
2	Backend / Logic	2	Flask Backend, Input Validation
3	Database / Storage	1	Dataset Storage
4	API / Integration	1	Flask API
5	Security / Authentication	0	Not Implemented